

Research Interests

Human–AI interaction, visual analytics, and reliable AI. I build interactive systems that help people diagnose model failures and guide how AI agents learn reusable skills, combining visual interfaces with behavioral evaluation.

Education

Nanjing University Sep 2023 – present
B.Eng. in Intelligent Software Engineering (expected Jun 2027)
School of Intelligent Software and Engineering
GPA: 88.4/100 | Top 10% in major.

Research Experience

HKUST (Guangzhou) Aug – Oct 2026
Visiting Research Student | Collaboration with Prof. Weikai Yang
Nanjing University, iMATE Lab Mar 2025 – present
Research Assistant | Advisor: Prof. Ke Xu
School of Intelligence Science and Technology; collaboration with Dr. Wenchao Li.

Recent Research

SliceTune: Supporting Interactive Intent-Guided Slice Finding for Model Validation

Submitted to CHI 2027

- An interactive workflow maps practitioners’ validation concerns to interpretable data attributes; a Feature Graph steers slice discovery and links failure evidence to repair planning.
- Evaluation includes a controlled study with 14 participants, three classification cases, and interviews with three experts.

SkillContour: Steering Agent Skill Consolidation with Behavioral Evidence

Submitted to CHI 2027

- A mixed-initiative workbench connects source patches, skill instructions, and execution behavior to support evidence-based diagnosis, revision, replay, and commitment.
- Evaluation combines a 14-participant study with deployment of participant-reviewed skill libraries on 45 held-out NBA tasks.

SkillCommit: Evolving Agent Skills through Behaviorally Validated Scope Expansion

Yu He, Weikai Yang.

Submitted to AAAI 2027 | arXiv preprint, 2026 | Paper

- Cross-instance replay and mechanism checks guide a hierarchical library of agent skills; source replay validates consolidated candidates before commitment.
- Experiments cover RuleArena, OpenExempt, and KOR-Bench, including transfer across model scales and families.

Publications and Preprints

NetworkCanvas: Supporting Progressive Network Visualization Exploration via Adaptive Recommendations

Wenchao Li, Yuewen Gao, **Yu He**, Cong Zhu, Ke Xu.

CHI 2026 | Published conference paper | DOI | PDF

Adaptive recommendations and a branching history of analytical states support progressive network exploration.

Research period: Mar – Sep 2025

- Implemented core visualization and feedback components; contributed to dataset integration, interaction design, and prototype refinement.
- Drove manuscript development, including conceptual framing, figure design, and experimental documentation.

MSSR: Memory-Aware Adaptive Replay for Continual LLM Fine-Tuning

Yiyang Lu, **Yu He**, Jianlong Chen, Hongyuan Zha.

arXiv preprint, 2026 | Paper

Sample memory estimates guide adaptive replay to mitigate forgetting during continual fine-tuning.

Research period: Aug 2025 – Jan 2026

- Led theoretical modeling of knowledge-retention dynamics and contributed to adaptive replay strategies.
- Designed and implemented replay scheduling; integrated datasets and pretrained models for comparative evaluation.

Additional Research

NewsTract — Multimodal Analysis of News Videos: Event-based Visual Summarization over Serial Broadcasts

Yuewen Gao, **Yu He**, Xianglei Lv, Yingying Dong, Ke Xu.

An interactive system for exploring event-level connections across news broadcasts.

Research period: Jun 2025 – Jan 2026

- Designed data models and visualization logic for event filtering, overview construction, and coordinated views.
- Contributed to manuscript development and user evaluation, including study design, results analysis, and technical figures.

Selected Honors and Awards

Nanjing University People's Scholarship	Nov 2025
Meritorious Winner, Mathematical Contest in Modeling Consortium for Mathematics and Its Applications (COMAP).	May 2025
National Third Prize, China Collegiate Innovation and Invention Competition	Dec 2024

Technical Skills

Programming: Python, TypeScript, JavaScript, Java, C++, SQL.

Machine learning: PyTorch; continual fine-tuning and agent evaluation.

Interactive systems: React, Vue 3, D3.js, G6.js, ECharts, Tailwind CSS, Element Plus.

Development: Git/GitHub, Spring Boot, Webpack.

Languages: Chinese (native); English.